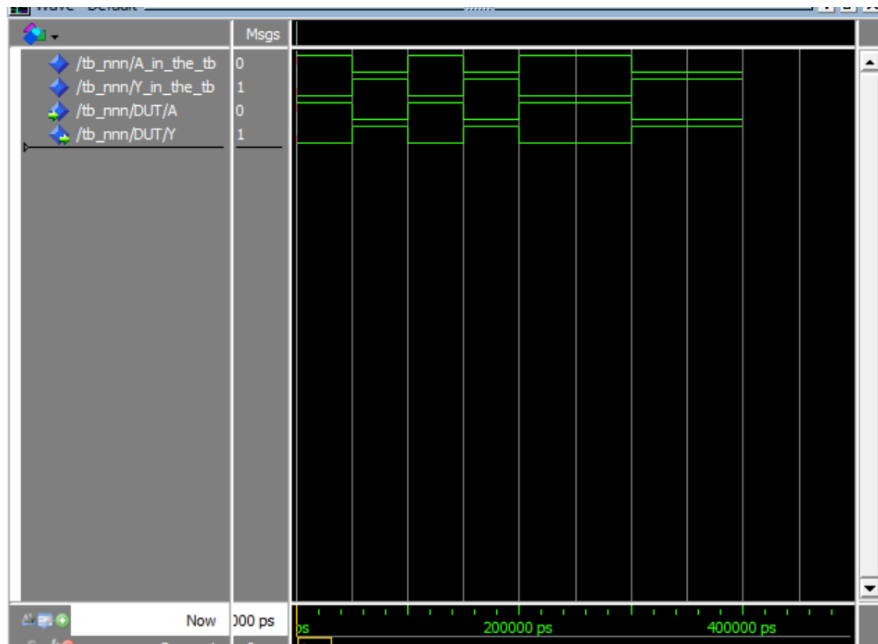
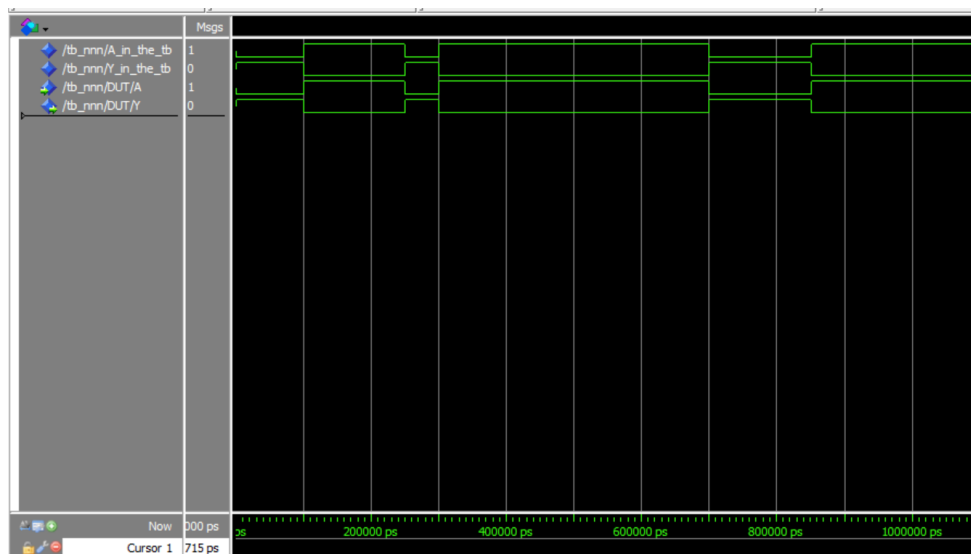


תרגיל הגשה 3

1. Testbench בשונה מ do הוא ממש חלק משפת VHDL למרות ששניהם בודקים את הרכיב ה Testbench מיועד ממש לצורכי בדיקה בשונה מ do שזה סתם אוסף של force .
- 2.



3.



.4

```
1 force a 1
2 run 50 ns
3 examine y
4 force a 0
5 run 50 ns
6 examine y
7 force a 1
8 run 50 ns
9 examine y
10 force a 0
11 run 50 ns
12 examine y
13 force a 1
14 run 100 ns
15 examine y
16 force a 0
17 run 100 ns
18 examine y
19
```

.5

```
1 force a 0
2 run 100 ns
3 examine y
4 force a 1
5 run 150 ns
6 examine y
7 force a 0
8 run 50 ns
9 examine y
10 force a 1
11 run 400 ns
12 examine y
13 force a 0
14 run 150 ns
15 examine y
16 force a 1
17 run 100 ns
18 examine y
19
```

```
# End time: 14:06:44 on Nov 20, 2024, Elapsed time: 0:29:15
# Errors: 0, Warnings: 4
# vsim -voptargs="+acc=lpn" tb_nnn
# Start time: 14:06:44 on Nov 20, 2024
# Loading std.standard
# Loading std.textio(body)
# Loading ieee.std_logic_1164(body)
# Loading ieee.std_logic_arith(body)
# Loading ieee.std_logic_unsigned(body)
# Loading ieee.std_logic_textio(body)
# Loading work.tb_nnn(yyy)
# Loading work.nnn(xxx)
add wave -r /*
VSI3M23> run 900 ns
# ** Warning: There is an 'U'|'X'|'W'|'Z'|'-' in an arithmetic operand, the result will be 'X'(es).
# Time: 0 ps Iteration: 0 Instance: /tb_nnn
# ** Warning: CONV_INTEGER: There is an 'U'|'X'|'W'|'Z'|'-' in an arithmetic operand, and it has been converted to 0
# Time: 0 ps Iteration: 0 Instance: /tb_nnn
# ** Warning: There is an 'U'|'X'|'W'|'Z'|'-' in an arithmetic operand, the result will be 'X'(es).
# Time: 0 ps Iteration: 0 Instance: /tb_nnn
# ** Warning: CONV_INTEGER: There is an 'U'|'X'|'W'|'Z'|'-' in an arithmetic operand, and it has been converted to 0
# Time: 0 ps Iteration: 0 Instance: /tb_nnn
# 0 ns | A is: 0 | Y is: 0
# 0 ns | A is: 1 | Y is: 0
# 50 ns | A is: 0 | Y is: 1
# 100 ns | A is: 1 | Y is: 0
# 150 ns | A is: 0 | Y is: 1
# 200 ns | A is: 1 | Y is: 0
# 300 ns | A is: 0 | Y is: 1
```

```
library IEEE;
use IEEE.std_logic_1164.all;
use IEEE.std_logic_unsigned.all;
use STD.textio.all;
use IEEE.std_logic_textio.all;

entity TB_NNN is
end TB_NNN;

architecture YYY of TB_NNN is

    component NNN
    port
    (
        A: in std_logic;
        Y: out std_logic
    );
    end component;

    signal A_in_the_tb, Y_in_the_tb: std_logic;

begin

    DUT: NNN
    port map
    (
        A => A_in_the_tb,
        Y => Y_in_the_tb
    );

    A_in_the_tb <= '1',
        '0' after 50 ns,
        '1' after 100 ns,
        '0' after 150 ns,
        '1' after 200 ns,
        '0' after 300 ns;

    L3: process(Y_in_the_tb)
        variable li: line;
    begin
        write(li, now);
        write(li, string'(" | "));
        write(li, string'(" A is: ") & integer'IMAGE(conv_integer('0' & A_in_the_tb)));
        write(li, string'(" | "));
        write(li, string'(" Y is: ") & integer'IMAGE(conv_integer('0' & Y_in_the_tb)));
        writeline(output, li);
    end process;
end YYY;
```

5E הקובץ הראשון זה הוספה של ספריות שנותנות את האפשרות להדפיס

הקובץ השני זה ההדפסה עצמה, שמדפיסה לנו את כל הקלט והפלט בכל זמן. וכך אנו יכולים לעקוב אחר הסיגנלים.